

A summary of periodic and random inspection policies

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ABSTRACT

Some systems work for a job with random working times. It would be useful for such systems that they are checked at the completion of working times to detect their failures, which is called a random inspection policy. Two inspection policies, where a system is checked at periodic or successive times and also at every completion of working times, are considered. The total expected costs until the detection of failure are obtained, and when the random working time is exponential, optimal inspection policies which minimize them are derived analytically. Furthermore, the inspection policy where the system is checked only at every completion of N th working time is proposed. Finally, as one of modified random inspection models, the backup model where the system goes back to the latest checking time when it has failed is taken up and analyzed, by using the inspection policy.

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1. Introduction

Most units in standby and in storage have to be checked at suitable times to detect their failures, which is called an *inspection policy*. Optimal policies which minimize the total expected cost until the detection of failures were derived [1]. Asymptotic inspection schedules and inspections policies for standby and storage units were discussed extensively [2, pp. 201–233], and their applications to gas pipelines and plants were shown [3, pp. 423–494]. The periodic and sequential inspection policies for a finite time span were summarized [4].

Some systems in offices and industries successively execute jobs and computer processes. For such systems, it would be impossible or impractical to maintain them in a strictly periodic fashion. For example, when a job has a variable working cycle and processing time, it would be better to do some maintenance after it has completed its work and process. The reliability quantities of the random age replacement policy were obtained analytically [1, pp. 72–74], using a renewal theory. When a unit is replaced only at random times, the properties of replacement times between two successive failed units were investigated [5]. The optimal replacement policies with random life cycle were discussed analytically, using a discount rate [6]. The various schedules of jobs which have random processing times were summarized [7].

We consider an operating system which executes a job with random working times Y_j ($j=1,2,\dots$) and $S_j \equiv \sum_{i=1}^j Y_i$, where $S_0 \equiv 0$ (Fig. 1). It is said that the system works for a random time Y_j . It is assumed that Y_j ($j=1,2,\dots$) is independent and has an identical distribution $G(x) \equiv \Pr\{Y_j \leq x\}$ with finite mean $1/\theta$. That is, the system works at a renewal process with its distribution $G(x)$. Then, the probability that the system works exactly j times in the interval $[0,x]$ is $G^{(j)}(x) - G^{(j+1)}(x)$, where $G^{(j)}(x)$ ($j=1,2,\dots$) denotes the j -fold Stieltjes convolution of $G(x)$ with itself and $G^{(0)}(x) \equiv 1$ for $x \geq 0$. In addition, $M(x) \equiv \sum_{j=1}^{\infty} G^{(j)}(x)$ which represents the expected number of works in $(0,x]$.

Suppose that the system deteriorates with its age, i.e., the total working time, irrespective of the number of works, and fails according to a general distribution $F(t)$ and its density function $f(t) \equiv dF(t)/dt$ with finite mean μ , where $\bar{\Phi}(t) \equiv 1 - \Phi(t)$ for any function $\Phi(t)$.

We apply the inspection policy to the system with a random working time Y_j : It is assumed in Section 2 that the system is checked at successive working times S_j and also at periodic times kT ($k=1,2,\dots$), and in Section 3 that it is checked at S_j and at successive times T_k ($k=1,2,\dots$). The total expected costs for two inspection policies are obtained, and optimal policies which minimize them are discussed analytically and computed numerically, when the random working time is exponential, where the optimal T^* of the periodic model was already derived [2, pp. 253–258, 8]. Furthermore, we propose one modified model where the downtime cost is suffered for the number of working jobs during system failure. It is assumed in Section 4 that the system is checked only at every completion of N th working time. When both working and failure times are exponential, an optimal

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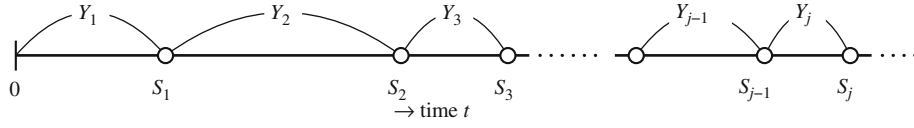


Fig. 1. Process of random working times.

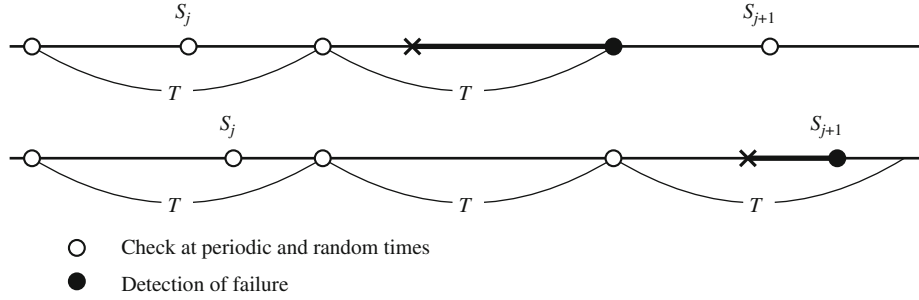


Fig. 2. Process of random and periodic inspections.

number N^* which minimizes the total expected cost is derived analytically. Furthermore, when the failure time has a Weibull distribution, N^* is computed numerically. It is shown in Conclusions that we can make similar discussions of obtaining an optimal policy which maximizes the availability [9,10].

The checkpoint models where the system goes back from its failure to the latest checkpoint time were proposed [11–13]. Finally, as one combined modification of the random inspection and checkpoint models, in Section 5 we take up the backup policy where we execute the backup operation until the latest checking

and the probability that it is detected by random check is

$$\sum_{k=0}^{\infty} \int_{kT}^{(k+1)T} \left\{ \sum_{j=0}^{\infty} \int_0^t [G((k+1)T-x) - G(t-x)] dG^{(j)}(x) \right\} dF(t), \quad (2)$$

where (1)+(2)=1.

Let c_P be the cost of periodic check, c_R be the cost of random check, and c_D be the downtime cost per unit of time for the time elapsed between a failure and its detection at the next check. Then, the total expected cost until failure detection is

$$\begin{aligned} C_1(T) &= \sum_{k=0}^{\infty} \int_{kT}^{(k+1)T} \left(\sum_{j=0}^{\infty} \{ (k+1)c_P + jc_R + c_D[(k+1)T-t] \} \int_0^t \bar{G}((k+1)T-x) dG^{(j)}(x) \right) dF(t) \\ &\quad + \sum_{k=0}^{\infty} \int_{kT}^{(k+1)T} \left(\sum_{j=0}^{\infty} \int_0^t \left\{ \int_{t-x}^{(k+1)T-x} [kc_P + (j+1)c_R + c_D(x+y-t)] dG(y) \right\} dG^{(j)}(x) \right) dF(t) \\ &= c_P \sum_{k=0}^{\infty} \bar{F}(kT) + c_R \int_0^{\infty} M(t) dF(t) - (c_P - c_R) \sum_{k=0}^{\infty} \int_{kT}^{(k+1)T} \left\{ G((k+1)T) - G(t) + \int_0^t [G((k+1)T-x) - G(t-x)] dM(x) \right\} dF(t) \\ &\quad + c_D \sum_{k=0}^{\infty} \int_{kT}^{(k+1)T} \left\{ \int_t^{(k+1)T} \bar{G}(y) dy + \int_0^t \left[\int_{t-x}^{(k+1)T-x} \bar{G}(y) dy \right] dM(x) \right\} dF(t). \end{aligned} \quad (3)$$

time, when the system is detected to have failed. Using the similar method of Section 4, the total expected cost until the completion of backup from a failure to the latest checking time is obtained, and an optimal policy which minimizes it is derived analytically.

2. Periodic inspection

Suppose that the system is checked at successive working times S_j ($j=1,2,\dots$) and also at periodic times kT ($k=1,2,\dots$) for a specified $T > 0$ (Fig. 2) [2, pp. 253–258, 8]. The failure is detected by either random or periodic checking times, whichever occurs first, and the process ends.

The probability that the failure is detected by periodic check is

$$\sum_{k=0}^{\infty} \int_{kT}^{(k+1)T} \left[\sum_{j=0}^{\infty} \int_0^t \bar{G}((k+1)T-x) dG^{(j)}(x) \right] dF(t), \quad (1)$$

In particular, when $T = \infty$, i.e., the system is checked only by random inspection, then the total expected cost is

$$\begin{aligned} C_1(\infty) &\equiv \lim_{T \rightarrow \infty} C_1(T) \\ &= c_R \int_0^{\infty} [1 + M(t)] dF(t) + c_D \left(\int_0^{\infty} F(t) \bar{G}(t) dt \right. \\ &\quad \left. + \int_0^{\infty} \left\{ \int_0^{\infty} [F(x+t) - F(x)] \bar{G}(t) dt \right\} dM(x) \right). \end{aligned} \quad (4)$$

Next, when $G(x) = 1 - e^{-\theta x}$, i.e., $M(x) = \theta x$, the total expected cost in (3) is

$$\begin{aligned} C_1(T) &= c_P \sum_{k=0}^{\infty} \bar{F}(kT) + c_R \theta \mu \\ &\quad + \left(c_R - c_P + \frac{c_D}{\theta} \right) \sum_{k=0}^{\infty} \int_{kT}^{(k+1)T} \{ 1 - e^{-\theta[(k+1)T-t]} \} dF(t). \end{aligned} \quad (5)$$

We find an optimal checking time T^* which minimizes $C_1(T)$. Differentiating $C_1(T)$ with respect to T and setting it equal to zero,

$$\frac{\sum_{k=0}^{\infty} (k+1) \int_{kT}^{(k+1)T} \theta e^{-\theta[(k+1)T-t]} dF(t)}{\sum_{k=0}^{\infty} k f(kT)} - (1 - e^{-\theta T}) = \frac{c_P}{c_R - c_P + c_D/\theta} \quad (6)$$

for $c_R + c_D/\theta > c_P$. This is a necessary condition that an optimal T^* minimizes $C_1(T)$.

In addition, when $F(t) = 1 - e^{-\lambda t}$ for $\lambda < \theta$, the total expected cost $C_1(T)$ in (5) is

$$C_1(T) = \frac{c_P}{1 - e^{-\lambda T}} + \frac{c_R \theta}{\lambda} + \left(c_R - c_P + \frac{c_D}{\theta} \right) \left(1 - \frac{\lambda}{\theta - \lambda} \frac{e^{-\lambda T} - e^{-\theta T}}{1 - e^{-\lambda T}} \right). \quad (7)$$

Clearly,

$$C_1(0) \equiv \lim_{T \rightarrow 0} C_1(T) = \infty, \quad C_1(\infty) = c_R \left(\frac{\theta}{\lambda} + 1 \right) + \frac{c_D}{\theta}.$$

Eq. (6) is simplified as

$$\frac{\theta}{\theta - \lambda} [1 - e^{-(\theta - \lambda)T}] - (1 - e^{-\theta T}) = \frac{c_P}{c_R - c_P + c_D/\theta}, \quad (8)$$

whose left-hand side is strictly increasing from 0 to $\lambda/(\theta - \lambda)$.

Therefore, if $\lambda/(\theta - \lambda) > c_P/(c_R - c_P + c_D/\theta)$, i.e., $c_R + c_D/\theta > (\theta/\lambda)c_P$, then there exists a finite and unique T^* ($0 < T^* < \infty$) which satisfies (8). The physical meaning of the condition $c_R + c_D/\theta > [(\lambda/\lambda)/(1/\theta)]c_P$ is that the total of checking cost and downtime cost of the mean interval between random checks is higher than the periodic cost for the expected number of random checks until failure detection. Conversely, if $c_R + c_D/\theta \leq (\theta/\lambda)c_P$, then the periodic inspection is not needed. If $\lambda > \theta$, then the left-hand side of (8) also increases strictly with T from 0 to ∞ , and hence, there exists a finite and unique T^* which satisfies (8). Note that if $\lambda = \theta$, then the left-hand side of (8) becomes $\theta T - (1 - e^{-\theta T})$.

Example 1. Suppose that the failure time has a Weibull distribution and the working time is exponential, i.e., $F(t) = 1 - \exp(-\lambda t^m)$ ($m \geq 1$) and $G(x) = 1 - e^{-\theta x}$. Then, from (6), an optimal checking time T^* satisfies

$$\frac{\sum_{k=0}^{\infty} (k+1) \int_{kT}^{(k+1)T} \theta e^{-\theta[(k+1)T-t]} \lambda m t^{m-1} e^{-\lambda t^m} dt}{\sum_{k=0}^{\infty} k \lambda m (kT)^{m-1} e^{-\lambda (kT)^m}} - (1 - e^{-\theta T}) = \frac{c_P}{c_R - c_P + c_D/\theta}. \quad (9)$$

In case of $m=1$, (9) agrees with (8). When $1/\theta$ tends to infinity, (9) becomes

$$\frac{\sum_{k=0}^{\infty} e^{-\lambda (kT)^m}}{\sum_{k=0}^{\infty} k \lambda m (kT)^{m-1} e^{-\lambda (kT)^m}} - T = \frac{c_P}{c_D}, \quad (10)$$

Table 1
Optimal time T^* and expected cost $C_1(T^*)/c_D$ when $1/\lambda = 100$, $c_P/c_D = 2$, and $c_R/c_D = 1$.

$1/\theta$	$m=1$		$m=2$		$m=3$	
	T^*	$C_1(T^*)/c_D$	T^*	$C_1(T^*)/c_D$	T^*	$C_1(T^*)/c_D$
1	∞	102.000	∞	10.862	∞	6.144
2	∞	53.000	23.680	7.423	7.017	4.854
3	∞	37.333	16.699	6.830	6.512	4.613
4	∞	30.000	14.011	6.748	6.303	4.551
5	∞	26.000	12.264	6.783	6.187	4.541
10	∞	21.000	8.081	6.914	5.969	4.589
15	49.941	22.165	7.183	6.937	5.898	4.630
20	32.240	22.210	6.819	6.945	5.861	4.657
25	27.842	22.120	6.621	6.948	5.839	4.675
50	22.568	21.799	6.266	6.953	5.794	4.716
∞	19.355	21.487	5.954	6.966	5.748	4.771

which corresponds to the periodic inspection with Weibull failure time [2, p. 204].

Table 1 presents the optimal time T^* and the resulting expected cost $C_1(T^*)/c_D$ for $m=1, 2, 3$ and $1/\theta = 1, 2, 3, 4, 5, 10, 15, 20, 25, 50, \infty$ when $1/\lambda = 100$, $c_P/c_D = 2$ and $c_R/c_D = 1$. This indicates that T^* decreases with $1/\theta$ and m . However, if the mean working time $1/\theta$ exceeds a threshold level, optimal values vary little for given m . Thus, it would be sufficient to check the unit at the smallest T^* for large $1/\theta$, which satisfies (10). It is of great interest that $C_1(T^*)$ has no monotonous property for $1/\theta$. This suggests that there might exist an optimal combined inspection policy in which the system is checked at periodic times kT and at every N th working time (see, Section 4).

Furthermore, we consider the inspection policy in which failures are detected only at periodic times kT . Then, because the inspection has no relation to any working time, this corresponds to the standard inspection policy and has been discussed [2, p. 202]. However, it would be appropriate to assume in this model that c_D is the cost of the number of jobs for which a failed system has worked before its detection, without the downtime cost per unit of time. Then, the total expected cost until failure detection is

$$\begin{aligned} \tilde{C}_1(T) &= \sum_{k=0}^{\infty} \int_{kT}^{(k+1)T} \{(k+1)c_P + c_D[M((k+1)T) - M(t)]\} dF(t) \\ &= \sum_{k=0}^{\infty} \{c_P + c_D[M((k+1)T) - M(kT)]\} \bar{F}(kT) - \int_0^{\infty} M(t) dF(t). \end{aligned} \quad (11)$$

In particular, when $G(t) = 1 - e^{-\theta t}$, $M(t) = \theta t$, and hence $\tilde{C}_1(T)$ agrees with the expected cost of a standard inspection policy [2, p. 203].

3. Sequential inspection

Suppose that the unit is checked at working times S_j and also at successive times T_k ($k=1, 2, \dots$), where $T_0=0$ (Fig. 3). Then, by the similar method of obtaining (3), the total expected cost is

$$\begin{aligned} C_2(\mathbf{T}) &= c_P \sum_{k=0}^{\infty} \bar{F}(T_k) + c_R \int_0^{\infty} M(t) dF(t) - (c_P - c_R) \\ &\quad \times \sum_{k=0}^{\infty} \int_{T_k}^{T_{k+1}} \left\{ G(T_{k+1}) - G(t) + \int_0^T [G(T_{k+1}-x) - G(t-x)] dM(x) \right\} dF(t) \\ &\quad + c_D \sum_{k=0}^{\infty} \int_{T_k}^{T_{k+1}} \left\{ \int_t^{T_{k+1}} \bar{G}(y) dy + \int_0^t \left[\int_{t-x}^{T_{k+1}-x} \bar{G}(y) dy \right] dM(x) \right\} dF(t), \end{aligned} \quad (12)$$

where $\mathbf{T} = \{T_1, T_2, \dots\}$. In particular, when $G(x) = 1 - e^{-\theta x}$,

$$\begin{aligned} C_2(\mathbf{T}) &= c_P \sum_{k=0}^{\infty} \bar{F}(T_k) + c_R \theta \mu \\ &\quad + \left(c_R - c_P + \frac{c_D}{\theta} \right) \sum_{k=0}^{\infty} \int_{T_k}^{T_{k+1}} [1 - e^{-\theta(T_{k+1}-t)}] dF(t). \end{aligned} \quad (13)$$

Differentiating $C_2(\mathbf{T})$ in (13) with T_k and setting it equal to zero,

$$1 - e^{-\theta(T_{k+1}-T_k)} = \frac{\int_{T_{k-1}}^{T_k} \theta e^{-\theta(T_k-t)} dF(t)}{f(T_k)} - \frac{c_P}{c_R - c_P + c_D/\theta} \quad (k=1, 2, \dots). \quad (14)$$

When $1/\theta$ tends to infinity, i.e., $\theta \rightarrow 0$, (14) becomes

$$T_{k+1} - T_k = \frac{F(T_k) - F(T_{k-1})}{f(T_k)} - \frac{c_P}{c_D},$$

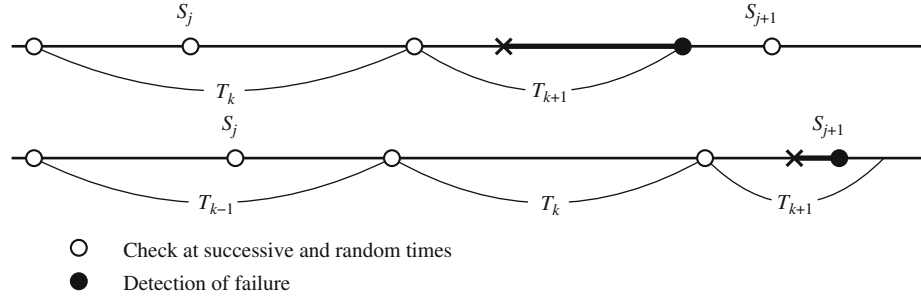


Fig. 3. Process of random and sequential inspections.

Table 2

Optimal schedule $\{T_k\}$ when $1/\lambda = 100$, $c_P/c_D = 2$, and $c_R/c_D = 1$.

k	$1/\theta$			
	10	50	∞	
1	9.85	8.63	8.36	
2	14.25	12.75	12.42	
3	17.81	16.12	15.75	
4	20.92	19.09	18.68	
5	23.75	21.79	21.35	
6	26.36	24.29	23.83	
7	28.80	26.63	26.16	
8	31.11	28.84	28.36	
9	33.30	30.90	30.46	
10	35.34	32.70	32.48	

which corresponds to the sequential inspection [1, p. 110, 2, p. 203]. Therefore, by using Algorithm 1 [1, p. 112, 2, p. 203], we can compute the optimal inspection schedule which satisfies (14).

Example 2. Suppose that the failure time has a Weibull distribution $(1 - e^{-\lambda t^2})$. Then, from (14),

$$1 - e^{-\theta(T_{k+1} - T_k)} = \frac{\int_{T_{k-1}}^{T_k} \theta e^{-\theta(T_k - t)} t e^{-\lambda t^2} dt}{T_k e^{-\lambda(T_k)^2}} - \frac{c_P}{c_R - c_P + c_D/\theta}.$$

As $1/\theta$ tends to infinity,

$$T_{k+1} - T_k = \frac{e^{-\lambda(T_{k-1})^2} - e^{-\lambda(T_k)^2}}{2\lambda T_k e^{-\lambda(T_k)^2}} - \frac{c_P}{c_D}.$$

Table 2 presents optimal T_k^* ($k = 1, 2, \dots, 10$) for $1/\theta = 10, 50, \infty$ when $1/\lambda = 100$, $c_P/c_D = 2$, and $c_R/c_D = 1$. This indicates that T_k^* decreases slowly with $1/\theta$, however, varies a little for $1/\theta$, and increases gradually with k . Compared to Table 1, it is of interest that $T_1^* > T^* > T_2^* - T_1^*$ for the same $1/\theta$.

4. Random inspection

Suppose that the system is checked at every N th working time, i.e., at the nN th number ($n = 1, 2, \dots$) of works. When $N = 1$, the unit is checked at every working time S_j ($j = 1, 2, \dots$). In this case,

$$\begin{aligned} & \int_0^\infty \left\{ \int_0^\infty [F(x+t) - F(x)] \bar{G}(t) dt \right\} dM(x) \\ &= \frac{1}{\theta} \int_0^\infty \bar{F}(x) dM(x) - \int_0^\infty \bar{F}(t) \left[\int_0^t \bar{G}(t-x) dM(x) \right] dt \\ &= \frac{1}{\theta} \int_0^\infty \bar{F}(t) dM(t) - \int_0^\infty \bar{F}(t) G(t) dt, \end{aligned}$$

and hence, from (4),

$$C_3(1) = C_1(\infty) = \left(c_R + \frac{c_D}{\theta} \right) \left[1 + \int_0^\infty \bar{F}(t) dM(t) \right] - \mu c_D.$$

When $N = 2$, i.e., the system is checked at every working time S_{2j} ($j = 1, 2, \dots$), the total expected cost is

$$\begin{aligned} C_3(2) &= c_R \sum_{j=0}^\infty (j+1) \int_0^\infty [G^{(2j)}(t) - G^{(2j+1)}(t)] dF(t) + c_D \left(\int_0^\infty F(t) [1 - G^{(2)}(t)] dt \right. \\ &\quad \left. + \int_0^\infty \left\{ \int_0^\infty [F(x+t) - F(x)] [1 - G^{(2)}(t)] dM^{(2)}(x) \right\} dt \right) \\ &= \left(c_R + \frac{2c_D}{\theta} \right) \left[1 + \int_0^\infty \bar{F}(t) dM^{(2)}(t) \right] - \mu c_D. \end{aligned}$$

Generally, when the unit is checked at the nN th ($n = 1, 2, \dots$) number of works,

$$C_3(N) = \left(c_R + \frac{Nc_D}{\theta} \right) \left[1 + \int_0^\infty \bar{F}(t) dM^{(N)}(t) \right] - \mu c_D \quad (N = 1, 2, \dots), \quad (15)$$

where $M^{(N)}(t) \equiv \sum_{n=1}^\infty G^{(nN)}(t)$ ($N = 1, 2, \dots$) and $M^{(1)}(t) \equiv M(t)$.

Example 3. When that $F(t) = 1 - e^{-\lambda t}$ and $G(t) = 1 - e^{-\theta t}$,

$$G^{(N)}(t) = 1 - \sum_{j=0}^{N-1} \frac{(\theta t)^j}{j!} e^{-\theta t}, \quad \int_0^\infty e^{-st} dG^{(N)}(t) = \left(\frac{\theta}{s+\theta} \right)^N.$$

Thus, because

$$\int_0^\infty e^{-st} dM^{(N)}(t) = \sum_{n=1}^\infty \left[\left(\frac{\theta}{s+\theta} \right)^N \right]^n = \frac{\left(\frac{\theta}{s+\theta} \right)^N}{1 - \left(\frac{\theta}{s+\theta} \right)^N},$$

the total expected cost is, from (15),

$$C_3(N) = \left(c_R + \frac{Nc_D}{\theta} \right) \frac{1}{1 - \left(\frac{\theta}{\lambda + \theta} \right)^N} - \frac{c_D}{\lambda} \quad (N = 1, 2, \dots). \quad (16)$$

From the inequality $C_3(N+1) - C_3(N) \geq 0$,

$$\sum_{j=1}^N \left(1 + \frac{\lambda}{\theta} \right)^j - N \geq \frac{c_R}{c_D/\theta},$$

i.e.,

$$\left(\frac{\lambda + \theta}{\theta} \right)^{N+1} - (N+1) \frac{\lambda}{\theta} - 1 \geq \frac{c_R}{c_D/\lambda} \quad (N = 1, 2, \dots), \quad (17)$$

whose left-hand side increases strictly with N to ∞ . Thus, there exists a finite and unique minimum N^* ($1 \leq N^* < \infty$) which satisfies (17). Clearly, if $\theta \leq \sqrt{\lambda c_D/c_R}$ then $N^* = 1$.

Furthermore, because

$$\left(1 + \frac{\lambda}{\theta} \right)^{N+1} \geq 1 + (N+1) \frac{\lambda}{\theta} + \frac{N(N+1)}{2} \left(\frac{\lambda}{\theta} \right)^2,$$

the upper number $\bar{N}$ of N^* is given by the unique minimum which satisfies

$$\frac{N(N+1)}{2} = \sum_{j=1}^N j \geq \frac{\theta^2 c_R}{\lambda c_D}. \quad (18)$$

Table 3
Optimal number N^* , expected cost $C_3(N^*)/c_D$ and upper number $\bar{N}$ when $1/\lambda = 100$.

$1/\theta$	$c_R/c_D=1$			$c_R/c_D=5$		
	N^*	$\bar{N}$	$C_3(N^*)/c_D$	N^*	$\bar{N}$	$C_3(N^*)/c_D$
1	14	14	15.352	30	32	35.618
2	7	7	15.884	15	16	36.195
3	5	5	16.456	10	11	36.769
4	4	4	17.083	8	8	37.388
5	3	3	17.507	6	6	37.912
10	1	2	21.000	3	3	40.740
15	1	1	22.667	2	2	43.527
20	1	1	26.000	2	2	47.273
25	1	1	30.000	1	1	50.000

Table 4
Optimal number N^* when $1/\lambda = 100$ and $c_R/c_D=5$.

$1/\theta$	$m=1$	$m=2$	$m=3$
1	30	9	6
2	15	5	3
3	10	3	2
4	8	2	2
5	6	2	2
10	3	1	1
15	2	1	1
20	2	1	1
25	1	1	1

Table 3 presents the optimal number N^* and the upper number $\bar{N}$ when $1/\lambda=100$. This indicates that optimal N^* decreases with $1/\theta$ and N^*/θ are almost the same for small $1/\theta$, and $\bar{N}$ gives very good approximation. Compared to Table 1 when $m=1$ and Table 3 when $c_R/c_D=1$, if $1/\theta$ is small, then $C_3(N^*)$ are smaller than $C_1(T^*)$. Note that when $1/\theta=10$, $C_1(T^*)=C_3(N^*)$. Clearly, we should adopt the periodic inspection when the mean working time is long. These tables suggest that if the working time is not so long, i.e., $1/\theta < 10$ in this case, the random inspection policy would be more and useful than the periodic one.

Next, suppose that the failure time has a Weibull distribution $F(t) = 1 - e^{-\lambda t^m}$ ($m \geq 1$). In this case, because the renewal density of $M^{(N)}(t)$ is, from [1, p. 57],

$$m^{(N)}(t) \equiv \frac{dM^{(N)}(t)}{dt} = \sum_{j=1}^{\infty} \frac{\theta(\theta t)^{Nj-1}}{(Nj-1)!} e^{-\theta t},$$

the expected cost in (15) is

$$C_3(N) = \left(c_R + \frac{Nc_D}{\theta} \right) \left[1 + \sum_{j=1}^{\infty} \int_0^{\infty} e^{-\lambda t^m} \frac{\theta(\theta t)^{Nj-1}}{(Nj-1)!} e^{-\theta t} dt \right] - \mu c_D. \quad (19)$$

Table 4 presents optimal N^* for $m=1, 2, 3$ and $1/\theta$ when $1/\lambda=100$ and $c_R/c_D=5$. When $m=1$, N^* is equal to Table 3 when $c_R/c_D=5$. Clearly, because the failure rate increases rapidly for $m=2, 3$, N^* become much smaller than those for $m=1$. When the working time is very long, the unit should be inspected at every working time.

5. Backup policy

We consider the case of failures being detected in the database and computer systems. In this case, a rollback operation has to be done until the last performed checkpoint and the system is rebuilt

from that point [11–14]. Combining the random inspection and checkpoint models, we propose the backup system where the system executes for a job with a random working time: Checking times are placed at the completion of N th working time (Fig. 4). When a failure occurs, it is detected immediately, and then the backup operation is executed until the latest checking time.

Suppose that the checking schedule is made at every completion of successive working times S_j ($j=1,2,\dots$). Let c_R be the cost required for each check. In addition, when a failure occurs at time t between S_j and S_{j+1} , we carry out the backup operation from failure to the latest checking time S_j ($=x$). This incurs a loss cost c_D per unit of time from S_j to failure; this includes all costs resulting from the working time and the rollback operation from failure to S_j , i.e., $c_D(t-x)$ (Fig. 4). The total expected cost until the completion of backup to the latest checking time when a failure occurs is, by the similar method of obtaining (3) when $c_p=0$,

$$C_4(1) = \sum_{j=0}^{\infty} \int_0^{\infty} \left\{ \int_0^t [jc_R + c_D(t-x)] \bar{G}(t-x) dG^{(j)}(x) \right\} dF(t) = c_R \int_0^{\infty} M(t) dF(t) + c_D \int_0^{\infty} \left[t \bar{G}(t) + \int_0^t (t-x) \bar{G}(t-x) dM(x) \right] dF(t).$$

Next, suppose that the system is checked only at every nN ($n=1,2,\dots$) working time. When $N=1$, the system is checked at every time of S_j ($j=1,2,\dots$). When $N=2$, i.e., the system is checked at the times of S_{2j} ($j=1,2,\dots$), the total expected cost is

$$C_4(2) = \sum_{j=0}^{\infty} \int_0^{\infty} \left\{ \int_0^t [jc_R + c_D(t-x)] [1 - G^{(2)}(t-x)] dG^{(2j)}(x) \right\} dF(t) = c_R \int_0^{\infty} M^{(2)}(t) dF(t) + c_D \int_0^{\infty} \left\{ t[1 - G^{(2)}(t)] + \int_0^t (t-x)[1 - G^{(2)}(t-x)] dM^{(2)}(x) \right\} dF(t).$$

Generally, when the system is checked at the nN ($n=1,2,\dots$) working time,

$$C_4(N) = c_R \int_0^{\infty} M^{(N)}(t) dF(t) + c_D \int_0^{\infty} \left\{ t[1 - G^{(N)}(t)] + \int_0^t (t-x)[1 - G^{(N)}(t-x)] dM^{(N)}(x) \right\} dF(t) \quad (N=1,2,\dots). \quad (20)$$



Fig. 4. Process of random checking time for a backup policy.

Table 5
Values of $\sum_{j=1}^N (1-A^j)$.

N	A								
	$\frac{1}{2}$	$\frac{2}{3}$	$\frac{3}{4}$	$\frac{4}{5}$	$\frac{5}{6}$	$\frac{6}{7}$	$\frac{7}{8}$	$\frac{8}{9}$	$\frac{9}{10}$
1	0.500	0.333	0.250	0.200	0.167	0.143	0.125	0.111	0.100
2	1.250	0.889	0.688	0.560	0.472	0.408	0.359	0.321	0.290
3	2.125	1.593	1.266	1.048	0.894	0.778	0.689	0.619	0.561
4	3.062	2.395	1.949	1.638	1.411	1.239	1.103	0.994	0.905
5	4.031	3.263	2.712	2.311	2.009	1.776	1.590	1.439	1.314
6	5.016	4.176	3.534	3.049	2.674	2.379	2.142	1.946	1.783
7	6.008	5.117	4.400	3.839	3.395	3.040	2.749	2.508	2.305
8	7.004	6.078	5.300	4.671	4.163	3.748	3.405	3.118	2.874
9	8.002	7.052	6.225	5.537	4.969	4.498	4.105	3.772	3.487
10	9.001	8.035	7.169	6.429	5.808	5.284	4.842	4.464	4.138

When $F(t) = 1 - e^{-\lambda t}$ and $G(t) = 1 - e^{-\theta t}$, from the results of Section 4, the total expected cost in (20) is

$$C_4(N) = \frac{c_R A^N}{1 - A^N} + \frac{c_D \lambda}{\theta(\theta + \lambda)} \frac{1}{1 - A^N} \sum_{j=1}^N j A^j \quad (N = 1, 2, \dots), \quad (21)$$

$$A(T) = \frac{\text{Mean time to failure}}{\text{Mean time to failure detection} + \text{total of checking times and maintenance time of a failed system}}$$

$$\begin{aligned} &= \mu \left/ \left(\beta_P \sum_{k=0}^{\infty} \bar{F}(kT) + \beta_R \int_0^{\infty} M(t) dF(t) - (\beta_P - \beta_R) \sum_{k=0}^{\infty} \int_{kT}^{(k+1)T} \left\{ G((k+1)T) - G(t) + \int_0^t [G((k+1)T - x) - G(t - x)] dM(x) \right\} dF(t) \right. \right. \\ &\quad \left. \left. + \sum_{k=0}^{\infty} \int_{kT}^{(k+1)T} \left\{ \int_t^{(k+1)T} \bar{G}(y) dy + \int_0^t \left[\int_{t-x}^{(k+1)T-x} \bar{G}(y) dy \right] dM(x) \right\} dF(t) + \mu + \beta_M \right) \right. \end{aligned} \quad (23)$$

where $A \equiv \theta/(\theta + \lambda)$. Clearly,

$$C_4(1) = \frac{c_R \theta}{\lambda} + \frac{c_D}{\theta + \lambda}, \quad C_4(\infty) = \frac{c_D}{\lambda}.$$

From the inequality $C_4(N+1) - C_4(N) \geq 0$,

$$N+1 - \frac{1 - A^{N+1}}{1 - A} \geq \frac{c_R}{c_D/(\theta + \lambda)},$$

i.e.,

$$\sum_{j=1}^N (1 - A^j) \geq \frac{c_R}{c_D/(\theta + \lambda)}, \quad (22)$$

whose left-hand side increases strictly to ∞ .

Therefore, there exists a finite and unique minimum N^* ($1 \leq N^* < \infty$) which satisfies (22). In particular, if $\lambda/(\theta + \lambda)^2 \geq c_R/c_D$ then $N^* = 1$.

Table 5 presents $\sum_{j=1}^N (1 - A^j)$ for A and N . For example, when $\theta/\lambda = 5$, i.e., the system fails one time per 5 works on the average, $A = 5/6$. In this case, when $D \equiv c_R/[c_D/(\theta + \lambda)]$, if $D \leq 0.167$ then $N^* = 1$, and if $0.167 < D \leq 0.472$ then $N^* = 2$, and so on. That is, the system should be checked at every completion of 2 working times. Clearly, if θ/λ increases then A increases, and hence, N^* increases.

6. Conclusions

We have discussed optimal inspection policies where the system is checked at suitable times and at random working times. As one modification of such models, we have shown the backup model where the system goes back from its failure to the latest checking time. Such methods would be applied to the checkpoint scheme for database and computer systems where a system consistency lost by some faults should be restored by some recovery techniques [12–14, pp. 123–147].

Calculations which compute optimal N^* in Sections 3 and 4 might be more troublesome except exponential cases. We should get some good idea for such computations because it is very difficult to obtain the renewal function $M(t)$ explicitly. However, it would be possible to compute optimal policies numerically by personal computers, as their functions have greatly developed.

Moreover, we have found in Table 1 that when the system is checked at periodic times kT and at every N th working time, there might exist optimal T^* and N^* which minimize the expected cost. This would offer interesting topics of optimization problems with two variables [14, pp. 149–186, 15] for further studies of random inspection policies.

The total expected cost has been adopted as an objective function in this paper. Let β_P be the time of one periodic check, β_R be the time of one random check, and β_M be the maintenance time of a failed system. Then, from (3), the availability for periodic inspection could be given by

Therefore, the policy maximizing $A(T)$ is the same one theoretically as minimizing $C_1(T)$ in (3) by replacing $c_i = \beta_i$ ($i = P, R$) and $c_D = 1$.

Both availability and cost are important reliability measures equivalently and are required simultaneously for most systems. Considering the circumstances and requirements of system operations, either or both of two measures would be choose definitely as an objective function. For example, we should consider mainly availabilities for aviation systems such as missiles, intercept fighters and radars because low availabilities may cause serious loss of their lives and facilities. On the other hand, we should consider mainly costs for commercial airplanes because airlines must compete each other seriously and make discount to win in the market. There have been issued many reports of inspection policies for actual systems [2, p. 202, 3]. Referring to such applied examples and modifying theoretical results, we should plan appropriate inspection schedules for objective systems.

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